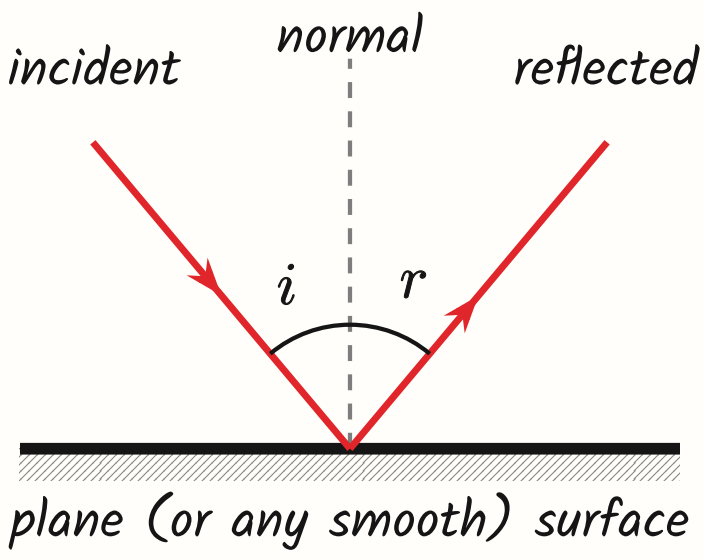
# Reflection of Light

**Class 12 · Cartesian sign convention, light from the LEFT · paraxial rays · thin lenses in air unless stated · examples: $n_{glass} = 1.5$, $n_{water} = 4/3$, $D = 25\,\text{cm}$**

## 1 Laws of reflection



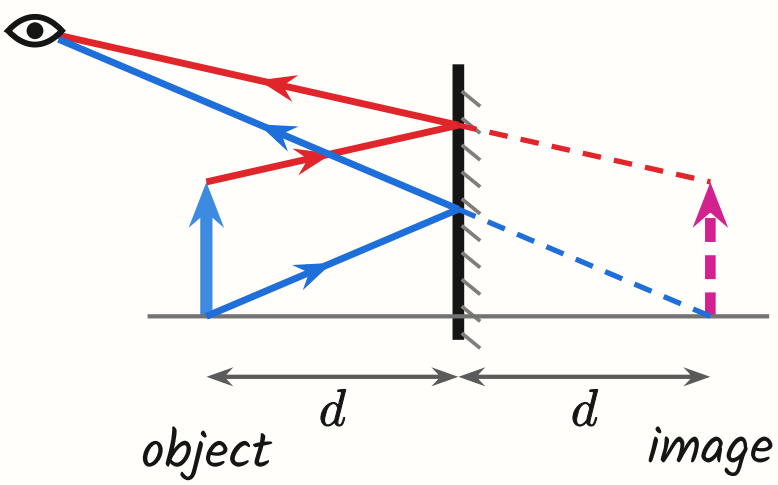

$$\angle i = \angle r$$

angles from the NORMAL · incident ray, normal, reflected ray in ONE plane · plane AND curved surfaces

$$\hat{e} = \hat{i} - 2(\hat{i} \cdot \hat{n})\,\hat{n}$$

$\hat{i}$, $\hat{e}$: along incident, reflected ray · $\hat{n}$: unit normal

## 2 Image in a plane mirror




- VIRTUAL, erect, same size, as far BEHIND as the object is in front ($m = +1$).
- Laterally inverted.
- PERFECT image with ALL rays (no paraxial condition).

## 3 Rectilinear propagation

- HOMOGENEOUS medium: light goes in straight lines; ray optics holds when apertures and obstacles $\gg \lambda$.

# Syllabus Checklist

JEE Main + JEE Advanced syllabus + standard results

## 1 JEE Main + JEE Advanced syllabus

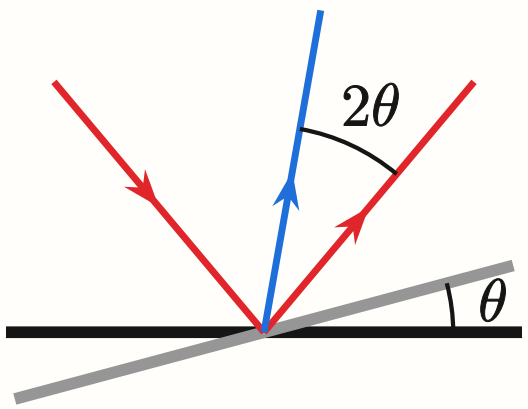

- ✓ Rectilinear propagation; reflection, plane mirrors — p1, p3
- ✓ Spherical mirrors: sign convention, mirror formula, $m$ — p4–p7
- ✓ Refraction, Snell's law, refractive index — p8
- ✓ Apparent depth, normal shift, lateral shift — p9, p10
- ✓ Total internal reflection, critical angle, optical fibre — p11, p12
- ✓ Refraction at a spherical surface — p13
- ✓ Thin lens: lensmaker's formula, lens formula, $m$ — p14–p16
- ✓ Power of a lens; lenses in contact and separated — p17
- ✓ Combinations of mirrors and thin lenses — p18
- ✓ Prism: deviation, minimum deviation, thin prism — p20, p21
- ✓ Dispersion, dispersive power — p22, p23
- ✓ Simple and compound microscope — p25, p26
- ✓ Astronomical (refracting) and reflecting telescopes — p27, p28


## 2 Also covered: extra topics + standard results


- ✓ Human eye and its defects (basics) — p24
- ✓ Inclined mirrors, rotating mirror, velocity of image — p3, p7, p16
- ✓ Newton's formula $x_1 x_2 = f^2$, $u$–$v$ graph — p7
- ✓ Lens in a liquid; different media on two sides — p14
- ✓ Silvered lens; displacement method; $4f$ minimum — p16, p18, p19
- ✓ Grazing incidence, no-emergence ($A > 2C$) — p21
- ✓ Achromatic, direct vision; scattering (basics) — p22, p23
- ✓ Terrestrial, Galilean, Newtonian and Cassegrain telescopes — p27, p28
- ✓ Traps, results table, solved questions — p29–p35

# Plane Mirror Results

## 1 Rotating the mirror




**reflected ray turns by $2\theta$**

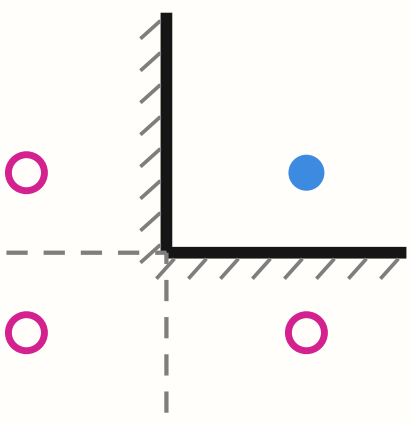
*incident ray FIXED, mirror (grey) turned by $\theta$*

$$\delta = 180^\circ - 2i$$

*deviation by ONE plane mirror*

## 2 Two inclined mirrors

$\theta = 90^\circ$: *3 images*



$$n = \frac{360^\circ}{\theta} - 1$$

*$360^\circ/\theta$ an EVEN integer · ODD: $n = \frac{360^\circ}{\theta}$ (object OFF the bisector) or $\frac{360^\circ}{\theta} - 1$ (ON it)*

**Example**

$\theta = 60^\circ$: $n = 6 - 1 = 5$. $\theta = 72^\circ$: 5 (off the bisector) or 4 (on it). Parallel: infinitely many.

## 3 Velocity of the image

$$v_{I\parallel} = v_{O\parallel}, \qquad v_{I\perp} = 2v_{M\perp} - v_{O\perp}$$

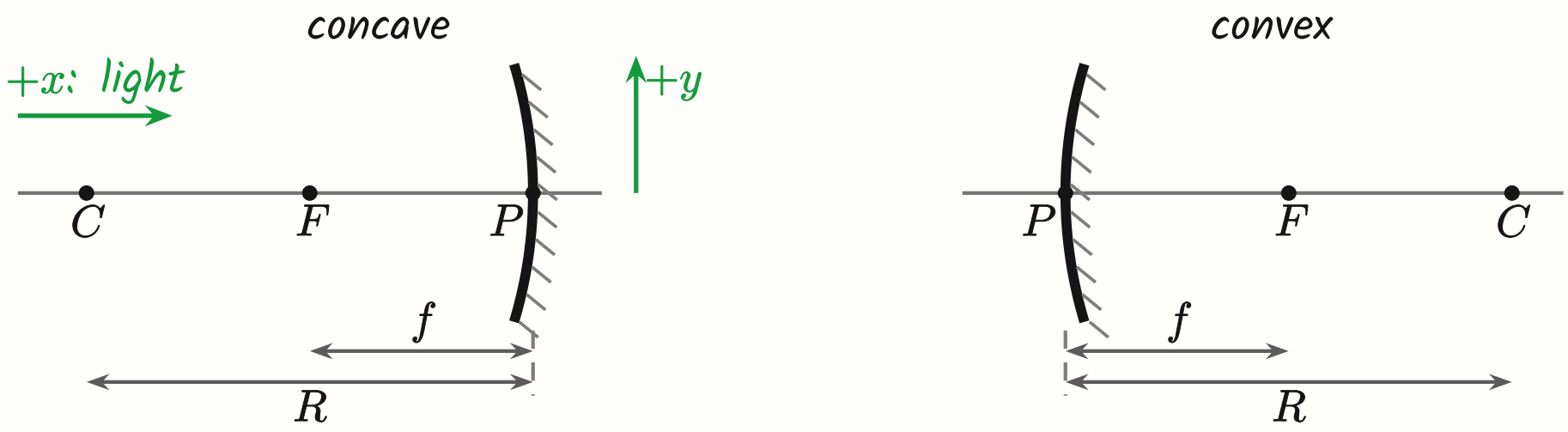
*plane mirror, GROUND frame · $\parallel$ along it, $\perp$ along its normal · object at rest: image at $2v_M$*

**Golden points**

★ *Full-length image (height $H$): mirror of $H/2$, any distance.*

# Spherical Mirrors

## 1 Terms and Cartesian sign convention




$$f = \frac{R}{2}$$

PARAXIAL rays · distances from $P$, + along the incident light, heights + up · concave $f, R < 0$; convex $f, R > 0$ · MIRROR: $f$ independent of the medium

## 2 Why only paraxial rays

$$CF' = \frac{R}{2\cos\theta} \rightarrow \frac{R}{2} \text{ only for small } \theta$$

parallel ray at height $h$, $\sin\theta = h/R$ · $F'$: where it crosses the axis

**Example**

$h = R/2$: $\theta = 30°$, $CF' = 0.577R$, $PF' = 0.423R$ (not $0.5R$): spherical aberration; a PARABOLIC mirror has none (for rays parallel to its axis).

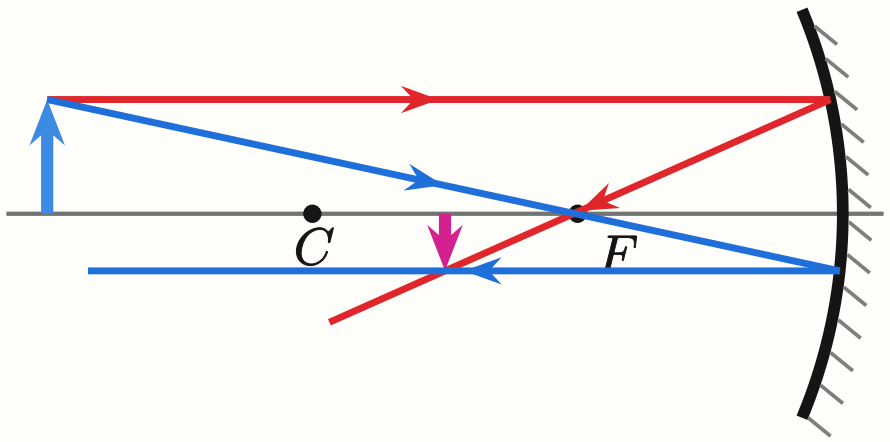
**Golden points**

★ Parallel → through $F$ (or as if from $F$); towards $F$ → parallel; towards $C$ → back on itself; at $P$ → symmetric (paraxial).

# Mirror Ray Diagrams

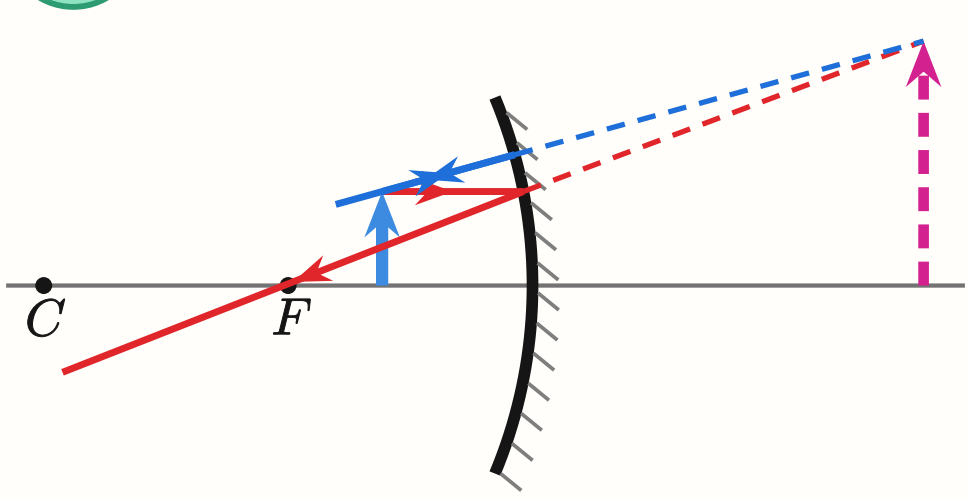
*paraxial rays; heights exaggerated for clarity*

## 1 Concave mirror, object beyond $C$



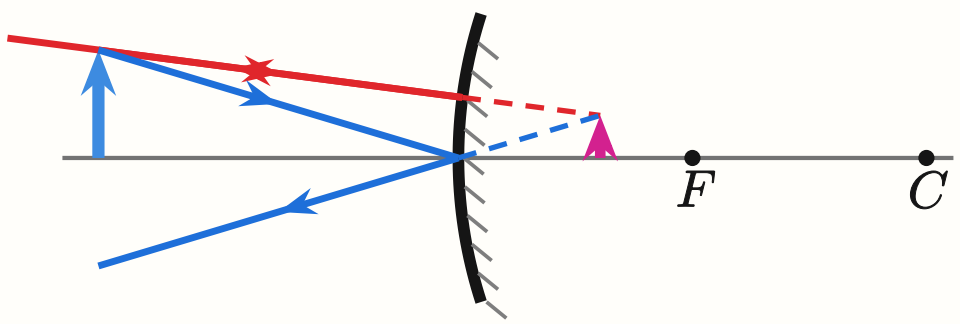

- Image between $F$ and $C$: REAL, inverted, diminished.
- Red: parallel $\rightarrow$ through $F$. Blue: through $F \rightarrow$ parallel.

## 2 Concave mirror, object between $F$ and $P$




- Image BEHIND the mirror: VIRTUAL, erect, magnified (shaving / make-up mirror).
- Blue: along $C \rightarrow$ object, returns on itself.

## 3 Convex mirror, any real object




- Image between $P$ and $F$ behind: VIRTUAL, erect, DIMINISHED (rear-view mirror: wide field).
- Red: towards $C$, back on itself. Blue: at $P$, symmetric.

# Mirror Formula

## 1 Mirror formula and magnification

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R} \qquad m=\frac{h'}{h}=-\frac{v}{u}$$

*PARAXIAL rays · Cartesian: from the pole, + along the incident light, heights + up · real object $u<0$ · concave and convex*

## 2 Where is the image? (concave mirror)

| *Object* | *Image* | *Nature* |
|---|---|---|
| *at* $\infty$ | *at* $F$ | *real, inverted, point size* |
| *beyond* $C$ | *between* $F$ *and* $C$ | *real, inverted, diminished* |
| *at* $C$ | *at* $C$ | *real, inverted, same size* |
| $C$ *to* $F$ | *beyond* $C$ | *real, inverted, magnified* |
| *at* $F$ | *at* $\infty$ | *real, inverted, very large* |
| $F$ *to* $P$ | *behind the mirror* | *VIRTUAL, erect, magnified* |
| *convex: any* | $P$ *to* $F$*, behind* | *VIRTUAL, erect, diminished* |

- *Single mirror, real object:* $m<0$ *real, inverted, in FRONT* ($v<0$)*;* $m>0$ *virtual, erect, BEHIND* ($v>0$).

**Example**

*Concave,* $|f|=10\,\text{cm}$*, object at* $30\,\text{cm}$*:* $\frac{1}{v}=-\frac{1}{10}+\frac{1}{30}=-\frac{1}{15}$*,* $v=-15\,\text{cm}$ *(real);* $m=-\frac{-15}{-30}=-0.5$.

# Image Velocity and Newton's Formula

## 1 Velocity of the image (spherical mirror)

$$v_I = -m^2\, v_O \quad \text{(along the axis)}$$

*object moving ALONG the axis, mirror at rest, paraxial · velocities along $+x$ · image moves OPPOSITE to the object*

$$v_{I\perp} = m\, v_{O\perp}$$

*object moving PERPENDICULAR to the axis at fixed $u$*

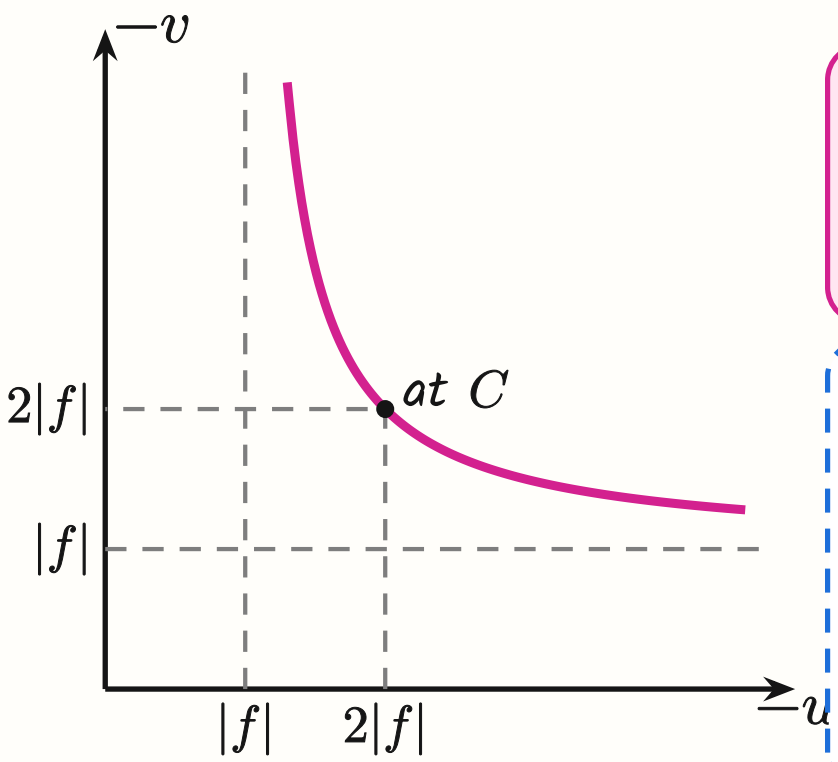
**Example**

*Concave, $|f| = 20\,\text{cm}$, object at $60\,\text{cm}$ approaching at $3\,\text{cm}\,\text{s}^{-1}$:* $v = -30\,\text{cm}$, $m = -0.5$, $v_I = -0.75\,\text{cm}\,\text{s}^{-1}$ *(moves AWAY).*

**Golden points**

★ *Plane mirror ($m = +1$, at rest): $v_I = -v_O$ along the normal. Object near $F$ ($|m|$ large): the image moves very fast.*

## 2 Newton's formula and the $u$–$v$ graph




*concave mirror, real images*

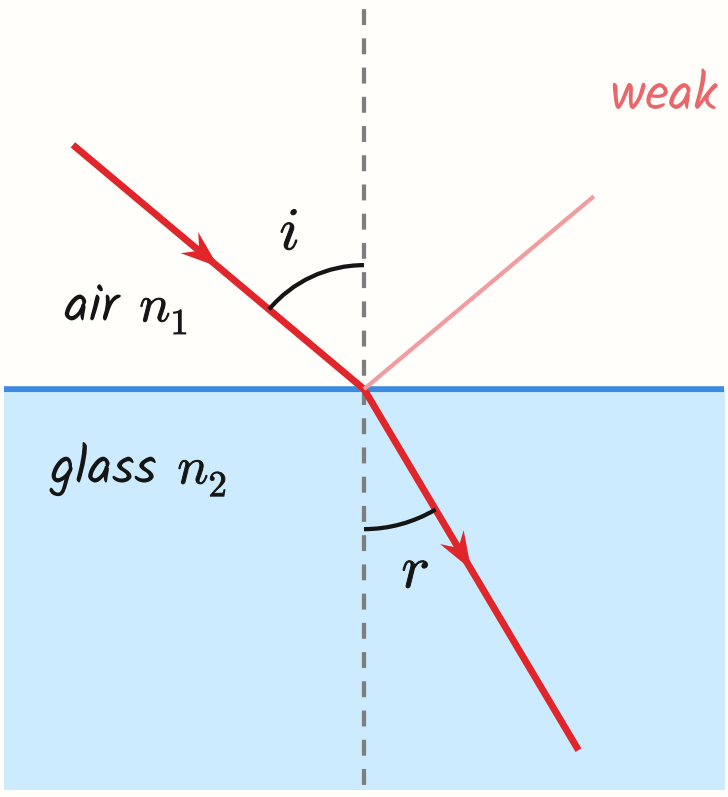
$$x_1 x_2 = f^2$$

*$x_1$, $x_2$: distances of object and image from the FOCUS · paraxial, same for a thin lens*

- *Graph: rectangular hyperbola, asymptotes at $|f|$, meets $v = u$ at $2|f|$.*
- $u = -30$, $f = -10$, $v = -15$: $x_1 = 20$, $x_2 = 5$, $x_1 x_2 = 100 = f^2$.

# Refraction and Snell's Law

## 1 Snell's law




$$n_1 \sin i = n_2 \sin r$$

$i$, $r$ from the NORMAL · incident ray, normal and refracted ray in one plane · $n = c/v$: absolute refractive index

**Example**

Air to glass ($n = 1.5$), $i = 50°$: $\sin r = 0.766/1.5$, $r = 30.7°$ (as drawn).

## 2 What changes, what does not

- FREQUENCY (colour) unchanged; $v = c/n$, $\lambda_m = \lambda_0/n$.
- Into a denser medium ($n_2 > n_1$): bends TOWARDS the normal; $i = 0$: no bending.
- Relative index $n_{21} = n_2/n_1 = v_1/v_2$; reversibility: $n_{12} = 1/n_{21}$, and $n_{12}\, n_{23}\, n_{31} = 1$.

**Example**

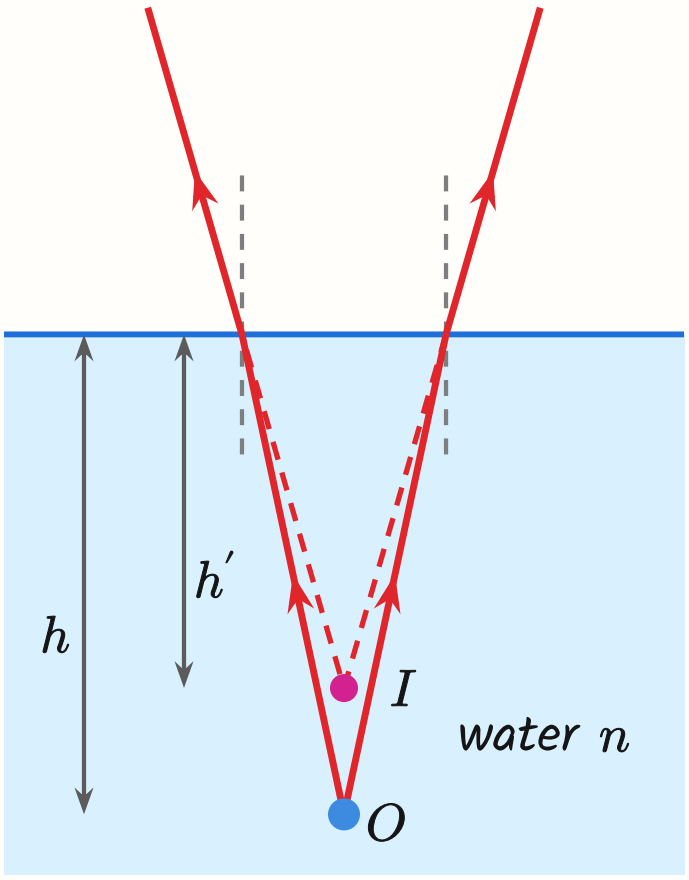
Glass, $n = 1.5$: $v = \frac{3\times10^8}{1.5} = 2 \times 10^8\,\mathrm{m\,s^{-1}}$; $600\,\mathrm{nm}$ (air) becomes $400\,\mathrm{nm}$ inside, frequency stays $5 \times 10^{14}\,\mathrm{Hz}$.

**Golden points**

★ Optical path $= nd$; time to cross thickness $d$ along the normal: $t = nd/c$.

# Apparent Depth and Normal Shift

## 1 Object under water, seen from above




$$h' = h\frac{n_{obs}}{n_{obj}} \Rightarrow h' = \frac{h}{n}$$

NEAR-NORMAL viewing · second form: object in medium $n$, observer in air

$$\Delta = h\left(1 - \frac{1}{n}\right)$$

rise of the image (near-normal viewing)

drawn exactly: $12°$ rays meet at $0.74h$; near-normal rays give $h/n = 0.75h$

## 2 Layers, slabs and the view from inside

$$h' = \frac{t_1}{n_1} + \frac{t_2}{n_2} + \cdots, \qquad \text{slab shift} = t\left(1 - \frac{1}{n}\right)$$

near-normal viewing from air · slab ($t$) between object and eye: image moves TOWARDS the eye

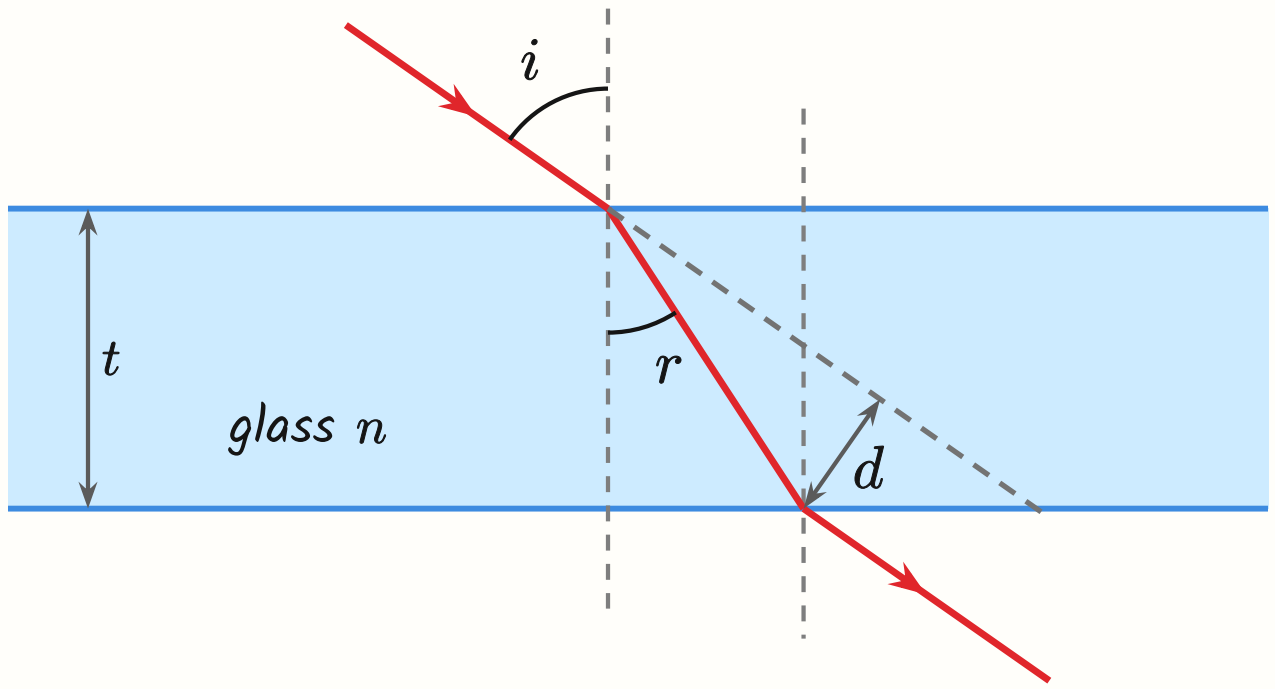
- Object in AIR seen from inside water (fish sees a bird): it looks FARTHER, $h' = nh$ (near-normal).

**Example**

$6\,\text{cm}$ glass $(1.5)$ on $8\,\text{cm}$ water $(4/3)$:
$h' = \frac{6}{1.5} + \frac{8}{4/3} = 4 + 6 = 10\,\text{cm}$ (real $14\,\text{cm}$). Slab $t = 6\,\text{cm}$, $n = 1.5$: shift $2\,\text{cm}$.

# Lateral Shift through a Slab

## 1 Parallel glass slab




$$d = \frac{t\,\sin(i - r)}{\cos r}$$

*PARALLEL-faced slab, SAME medium on both sides · emergent ray ∥ incident · $d$: perpendicular gap*

## 2 Limits

- $i = 0$: $d = 0$. *Grazing* ($i \to 90°$): $d \to t$.
- *Small* $i$ *(radians)*: $d \approx t\,i\left(1 - \frac{1}{n}\right)$.

**Example**

$t = 3\,\text{cm}$, $n = \sqrt{3}$, $i = 60°$: $\sin r = \frac{\sin 60°}{\sqrt{3}} = \frac{1}{2}$, $r = 30°$;

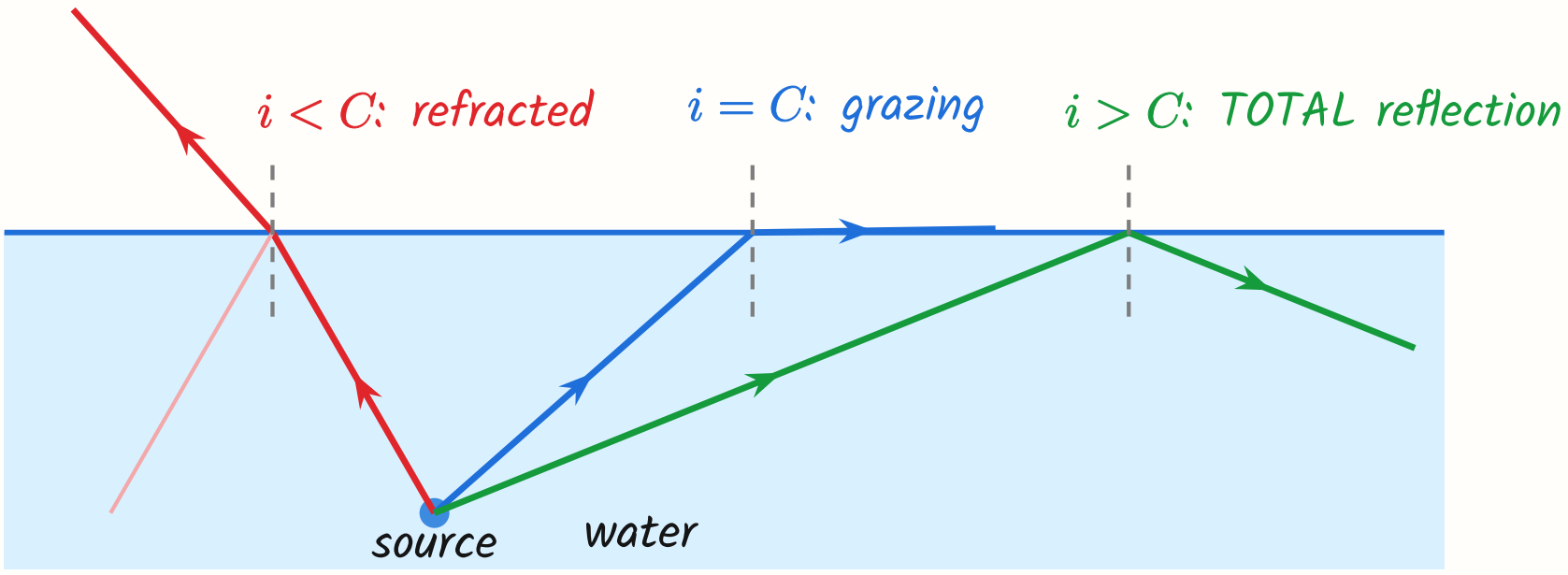
$d = \frac{3\sin 30°}{\cos 30°} = \sqrt{3} \approx 1.73\,\text{cm}$.

**Golden points**

★ *LATERAL shift (oblique ray, sideways) is not the NORMAL shift* $t(1 - 1/n)$ *(image seen through a slab, near-normal).*

# Total Internal Reflection

## 1 From denser to rarer




$$\sin C = \frac{n_{rarer}}{n_{denser}} \quad \left(= \frac{1}{n} \text{ into air}\right)$$

TIR only when light goes from DENSER to RARER and $i > C$ · 100 % reflected (no loss, unlike a metal mirror)

**Example**

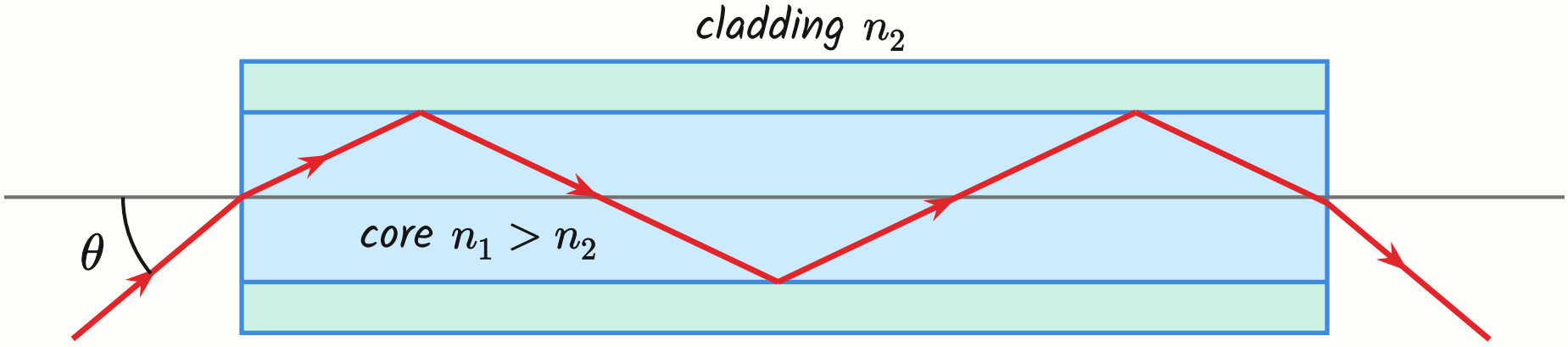
Critical angle: water $(4/3)$ $48.6°$; glass $(1.5)$ $41.8°$; diamond $(2.42)$ $24.4°$.

## 2 Uses

- Diamond sparkles: small $C$, cut so that light totally reflects many times.
- Mirage: hot air near the road has smaller $n$; light from the sky bends up and is totally reflected at grazing angles.
- Totally reflecting prism $(45°–90°–45°)$ turns rays by $90°$ or $180°$; needs $C < 45°$, i.e. $n > \sqrt{2}$ (periscopes, binoculars).

# Optical Fibre and Circle of Light

## 1 Optical fibre



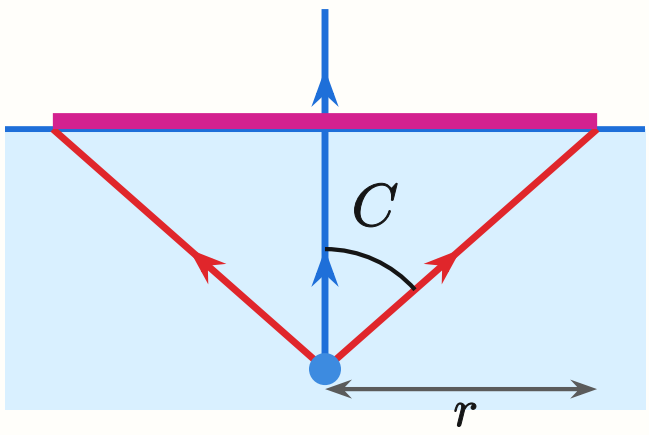

$$\sin\theta_a = \sqrt{n_1^2 - n_2^2} \quad \text{(numerical aperture)}$$

ray enters the end face from AIR; TIR at the core–cladding wall needs $\theta \le \theta_a$ · from a medium $n_0$: $n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ · if $\sqrt{n_1^2 - n_2^2} \ge n_0$ every ray is guided

**Example**

$n_1 = 1.5$, $n_2 = 1.2$: $\sqrt{2.25 - 1.44} = 0.9$, $\theta_a = 64.2°$ (drawn ray: $\theta = 40°$).

## 2 Circle of light




$$r = h\tan C = \frac{h}{\sqrt{n^2 - 1}}$$

point source at depth $h$ in a liquid of index $n$ under air: light leaves only through this circle

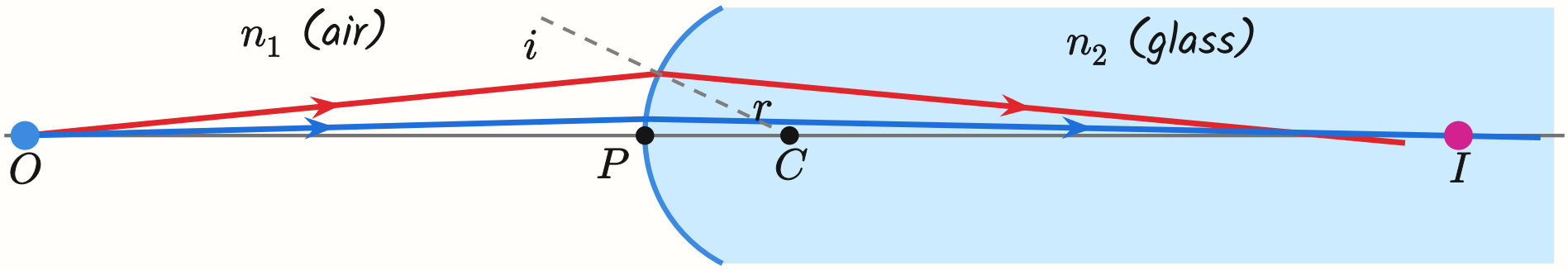
**Example**

Water, $h = 4\,\text{m}$:

$$r = \frac{4}{\sqrt{16/9 - 1}} = \frac{12}{\sqrt{7}} \approx 4.54\,\text{m}.$$

# Refraction at a Spherical Surface

## 1 Single curved surface




$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}, \qquad m = \frac{n_1 v}{n_2 u}$$

*PARAXIAL rays (rays traced exactly: the near-axis blue ray crosses almost at $I$, the wide red ray falls short) · Cartesian from the vertex $P$ · light from $n_1$ into $n_2$ · $R > 0$ if $C$ is on the outgoing side*

**Example**

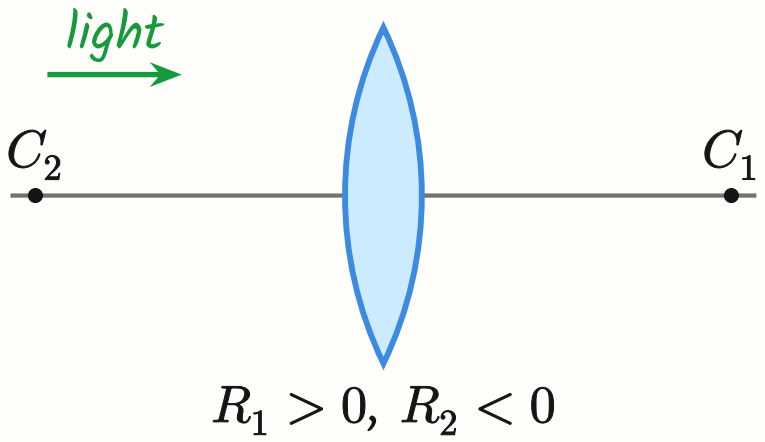
*Object in air $40\,\text{cm}$ from convex glass $(1.5)$, $R = 10\,\text{cm}$:* $\frac{1.5}{v} = \frac{0.5}{10} - \frac{1}{40}$, $v = 60\,\text{cm}$; $m = \frac{60}{1.5\times(-40)} = -1$.

## 2 Special cases

- *Plane surface ($R \to \infty$):* $v = \frac{n_2}{n_1}u$, *the apparent-depth result (p9).*
- *Power of the surface* $P = \frac{n_2 - n_1}{R}$; *focal lengths* $f_1 = -\frac{n_1 R}{n_2 - n_1}$, $f_2 = \frac{n_2 R}{n_2 - n_1}$.
- *Also for concave surfaces ($R < 0$) and glass to air ($n_1$: medium of the INCIDENT light).*

# Thin Lens: Lensmaker's Formula

## 1 Lensmaker's formula




$$\frac{1}{f} = (n_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$n_{rel} = n_{lens}/n_{medium}$ · THIN lens, paraxial, SAME medium both sides · $R_1$: surface met first, $R > 0$ if centre on the outgoing side

- Equi-biconvex: $f = \frac{R}{2(n-1)}$ $(= R$ for $n = 1.5)$. Plano-convex: $f = \frac{R}{n-1}$ $(= 2R)$.

## 2 Lens in a liquid

$$\frac{f_{liquid}}{f_{air}} = \frac{n - 1}{n/n_l - 1}$$

same thin lens (index $n$) in air and in a liquid $n_l$

- Glass $(1.5)$ in water $(4/3)$: $\frac{0.5}{1.125-1} = 4$, so $f_w = 4f_a$.
- $n_l > n$: convex lens DIVERGES (air bubble in water); $n_l = n$: $f \to \infty$.

## 3 Different media on the two sides

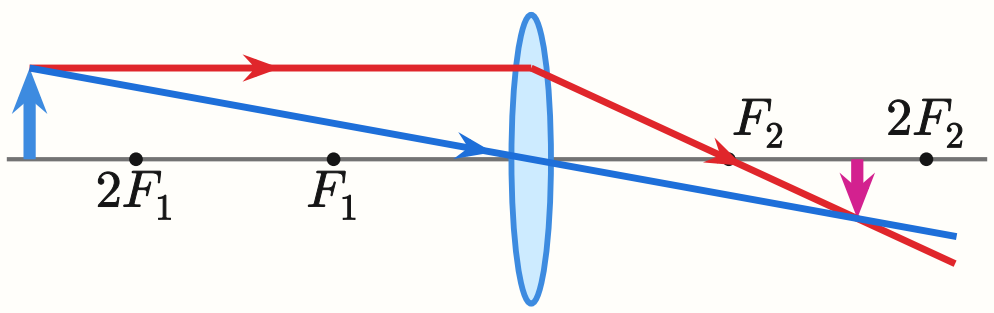
$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}$$

thin lens $n_2$, object side $n_1$, image side $n_3$, paraxial · $f_2/f_1 = -n_3/n_1$

# Lens Ray Diagrams

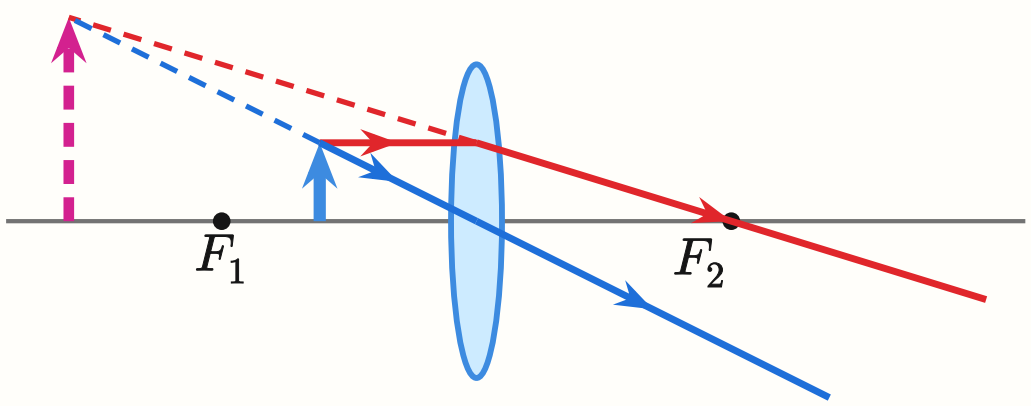
*thin lenses, paraxial rays; heights exaggerated for clarity*

## 1 Convex lens, object beyond $2F$



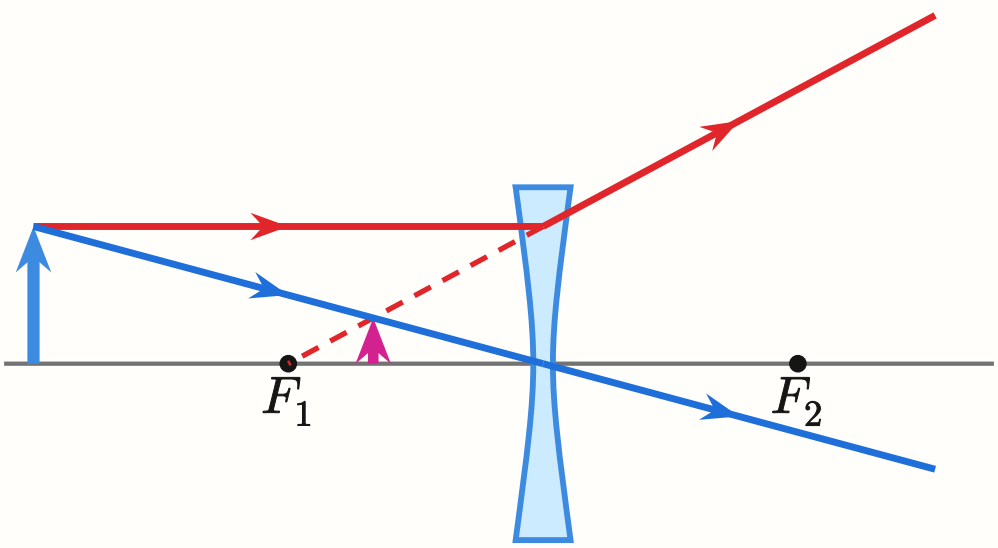

- Image between $F_2$ and $2F_2$: REAL, inverted, diminished (camera).
- Red: parallel → through $F_2$. Blue: through the optical centre, undeviated.

## 2 Convex lens, object inside $F$




- Image on the SAME side: VIRTUAL, erect, magnified (magnifying glass).
- Dashed: backward extensions meet at the image.

## 3 Concave lens, any real object




- Image between $F_1$ and the lens: VIRTUAL, erect, DIMINISHED.
- Red: parallel → diverges as if from $F_1$.

# Lens Formula

## 1 Lens formula and magnification

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \qquad m=\frac{h'}{h}=\frac{v}{u}$$

*THIN lens, paraxial, same medium both sides · Cartesian, from the optical centre, real object $u<0$ · convex $f>0$, concave $f<0$*

## 2 Where is the image? (convex lens)

| Object | Image | Nature |
|---|---|---|
| at $\infty$ | at $F_2$ | real, inverted, point size |
| beyond $2F_1$ | $F_2$ to $2F_2$ | real, inverted, diminished |
| at $2F_1$ | at $2F_2$ | real, inverted, same size |
| $F_1$ to $2F_1$ | beyond $2F_2$ | real, inverted, magnified |
| at $F_1$ | at $\infty$ | real, inverted, very large |
| $F_1$ to $O$ | object side | VIRTUAL, erect, magnified |
| concave: any | $F_1$ to $O$ | VIRTUAL, erect, diminished |

## 3 Two standard results

$$D_{min}=4f \qquad v_I=m^2 v_O$$

*$4f$: least real-object to REAL-image distance (convex lens, $u=-2f$) · $m^2$: object moving ALONG the axis, lens at rest, image moves the SAME way*

**Example**

$f=20\,\text{cm}$, object at $30\,\text{cm}$: $v=60\,\text{cm}$, $m=-2$; object approaching at $1\,\text{cm}\,\text{s}^{-1}$: image moves AWAY at $4\,\text{cm}\,\text{s}^{-1}$.

# Power and Lens Combinations

## 1 Power of a lens

$$P = \frac{1}{f\,(\text{in } m)} \qquad \text{dioptre: } 1\,\mathrm{D} = 1\,\mathrm{m}^{-1}$$

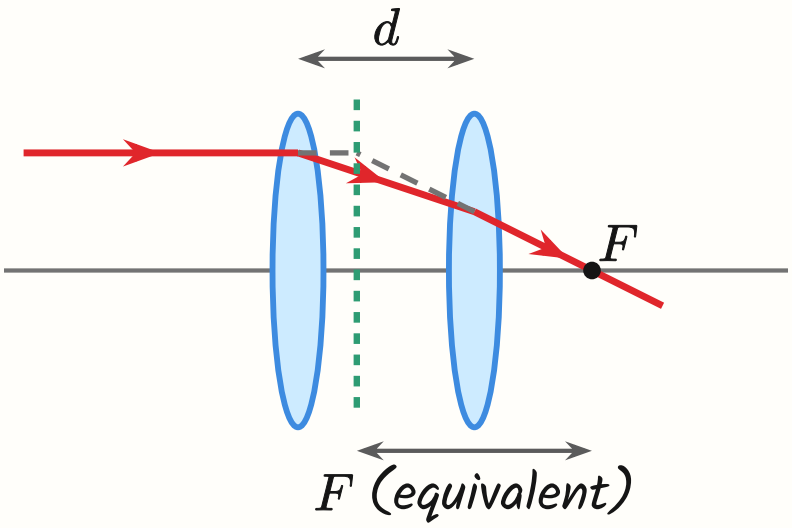
lens in AIR · convex (converging) $P > 0$, concave (diverging) $P < 0$

## 2 Thin lenses in contact

$$P = P_1 + P_2 + \cdots, \qquad m = m_1 m_2 \cdots$$

THIN lenses in CONTACT, coaxial · $1/F = 1/f_1 + 1/f_2 + \cdots$

## 3 Separated by a distance $d$




$$P = P_1 + P_2 - d\,P_1 P_2$$

two coaxial THIN lenses in air · $F$ is measured from the EQUIVALENT lens (green), not from either lens

**Example**

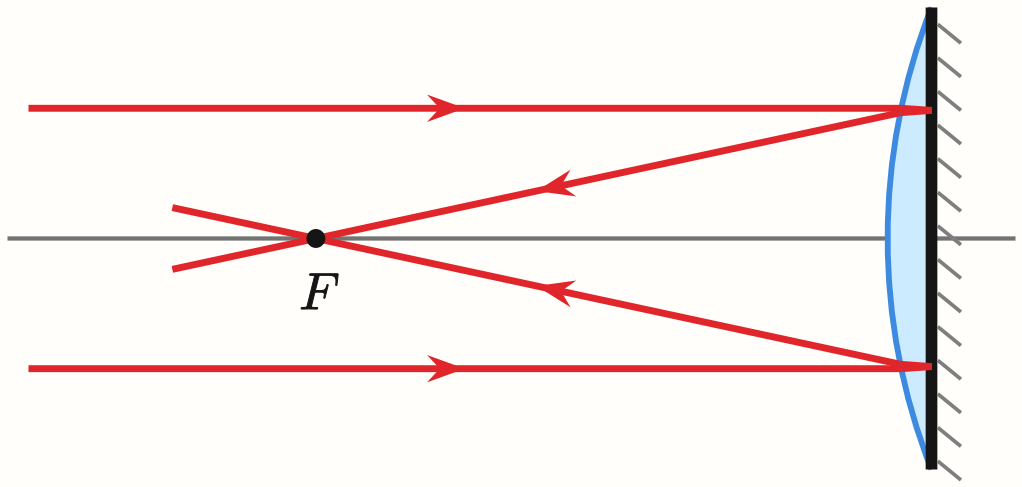
$f_1 = f_2 = 10\,\mathrm{cm}$, $d = 5\,\mathrm{cm}$:
$\frac{1}{F} = \frac{1}{10} + \frac{1}{10} - \frac{5}{100} = 0.15$,
$F = 6.67\,\mathrm{cm}$.

## 4 Cutting a lens

- Cut ALONG the axis: each half keeps $f$, full image, less light.
- Equi-biconvex cut PERPENDICULAR to the axis: two plano-convex lenses of $2f$ each.

# Silvered Lens and Lens–Mirror Systems

## 1 Silvered lens = a mirror




*plano-convex lens, PLANE face silvered (rays traced exactly)*

$$P_{eq} = 2P_L + P_M, \qquad f_{eq} = -\frac{1}{P_{eq}}$$

*thin lens silvered on one face, paraxial: light crosses the lens TWICE ·* $P_L = 1/f_L$, $P_M = -1/f_M$ *(concave* $P_M > 0$*) ·* $P_{eq} > 0$*: CONCAVE mirror*

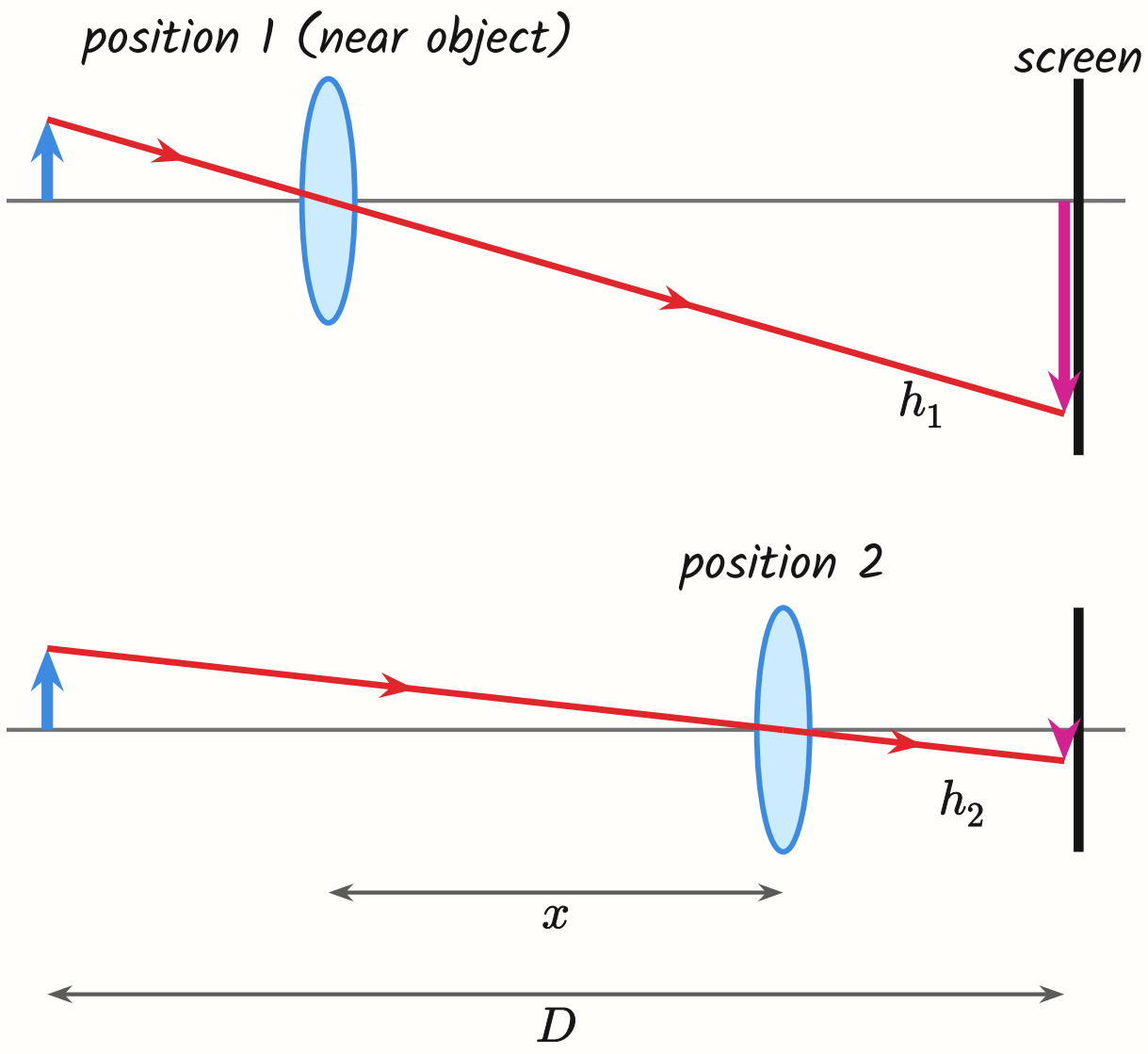
- *Plano-convex: PLANE face silvered* $f = \frac{R}{2(n-1)}$*; CURVED face silvered* $f = \frac{R}{2n}$*.* ($n = 1.5$, $R = 20\,\text{cm}$: $20\,\text{cm}$ *and* $\frac{20}{3} \approx 6.67\,\text{cm}$.)

## 2 Lens and mirror apart

- *Image by the lens* $\rightarrow$ *object for the mirror (may be virtual)* $\rightarrow$ *image by the mirror* $\rightarrow$ *back through the lens with light going RIGHT to LEFT (re-apply the convention from that side).* $m = m_1 m_2 m_3$.
- *Object at* $F$ *of a convex lens, plane mirror behind: light returns onto the object (finds* $f$*).*

# Displacement Method

## 1 Two lens positions, one screen




$$f = \frac{D^2 - x^2}{4D}, \qquad h = \sqrt{h_1 h_2}, \qquad m_1 m_2 = 1$$

*convex lens, object–screen distance $D$ FIXED and $D > 4f$ · $x$: distance between the two lens positions ($u$ and $v$ swap) · $D = 4f$: only one position*

**Example**

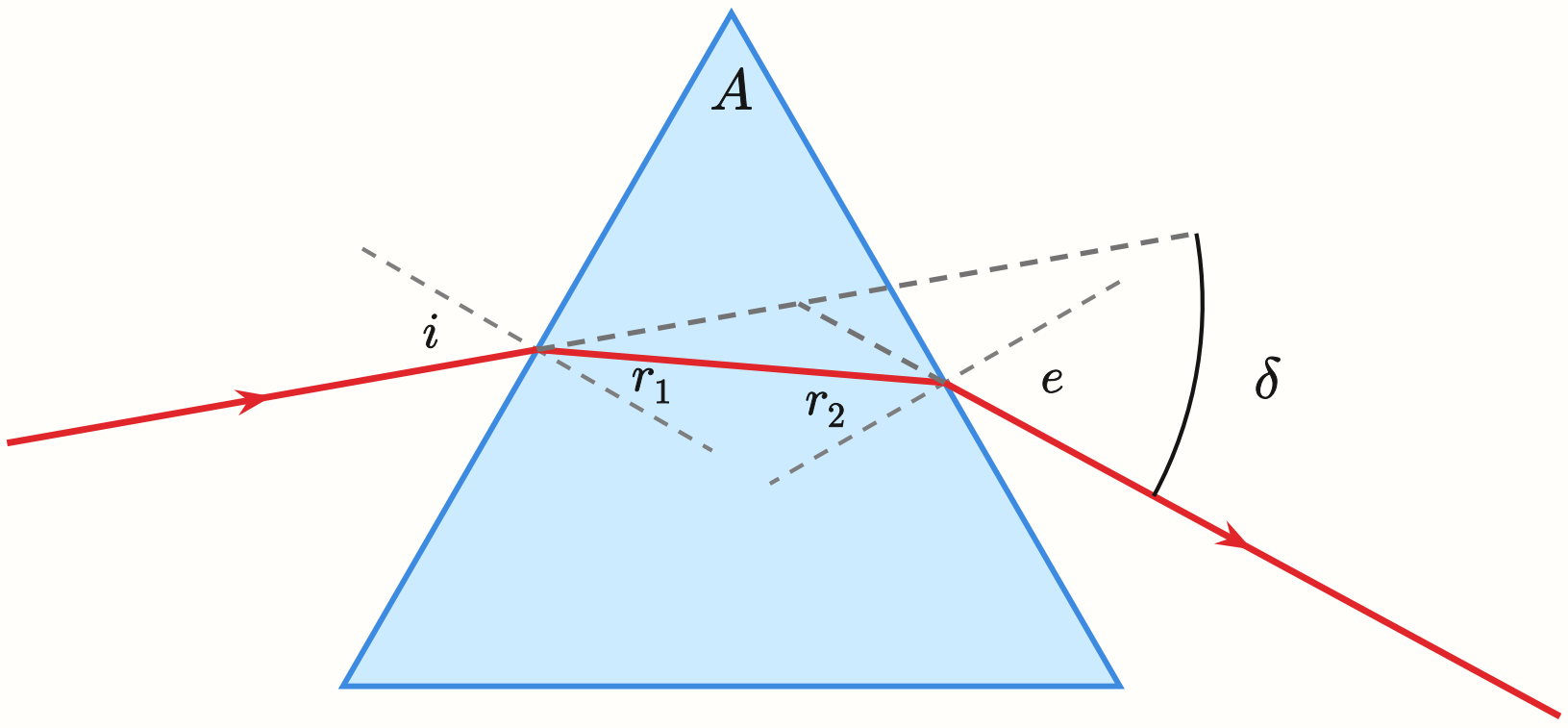
$D = 90\,\text{cm}$, $x = 30\,\text{cm}$: $f = \frac{8100-900}{360} = 20\,\text{cm}$. *Images* $8\,\text{cm}$ *and* $2\,\text{cm}$: *object* $\sqrt{16} = 4\,\text{cm}$.

**Golden points**

★ $D < 4f$: *no real image on the screen for any lens position. Both images are real and inverted,* $h_1 h_2 = h^2$.

# Prism: Deviation

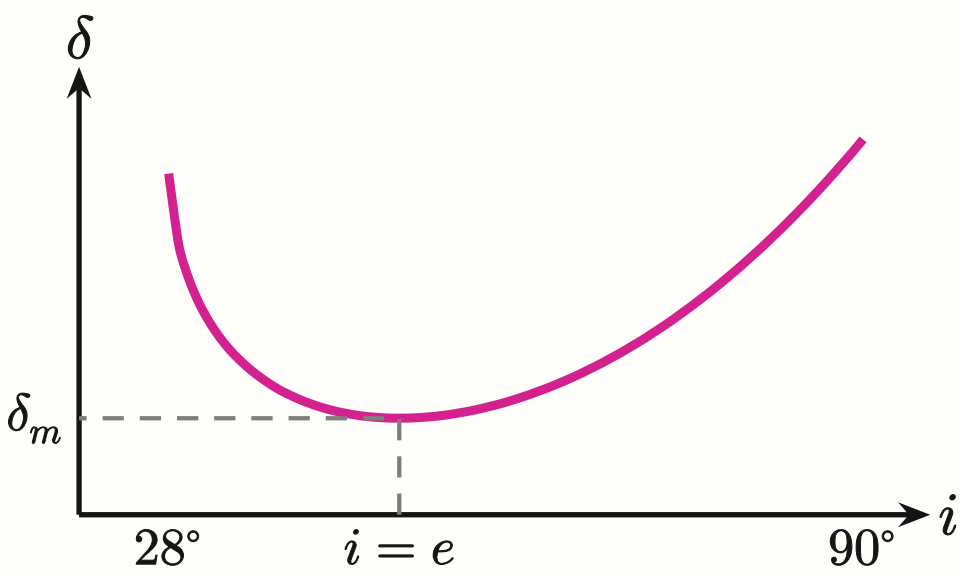
## 1 Path through a prism




$$r_1 + r_2 = A, \qquad \delta = i + e - A$$

*prism of angle $A$ in air, ray in a principal section · drawn exactly: $A = 60°$, $n = 1.5$, $i = 40°$ gives $e = 58.5°$, $\delta = 38.5°$*

## 2 Deviation against incidence




- *Drawn for $A = 60°$, $n = 1.5$: one MINIMUM $\delta_m = 37.2°$ at $i = e = 48.6°$.*
- *Same $\delta$ for $i$ and $e$ interchanged (reversibility), so the graph is not symmetric in $i$ but each $\delta$ with $\delta_m < \delta \leq \delta_{max}$ (57.9° here, at $i = 27.9°$ and at $i = 90°$) occurs twice.*

# Minimum Deviation and Thin Prism

## 1 Minimum deviation

$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

*prism in air · at $\delta_m$: $i = e$, $r_1 = r_2 = A/2$, ray inside ∥ base (isosceles prism)*

**Example**

*Equilateral, $n = 1.5$: $\sin\frac{60°+\delta_m}{2} = 0.75$, so $\frac{60°+\delta_m}{2} = 48.6°$, $\delta_m = 37.2°$.*

## 2 Thin prism

$$\delta = (n-1)A$$

*SMALL $A$ AND small (near-normal) $i$ · then independent of $i$ · $A = 6°$, $n = 1.5$: $\delta = 3°$*

## 3 Limiting cases

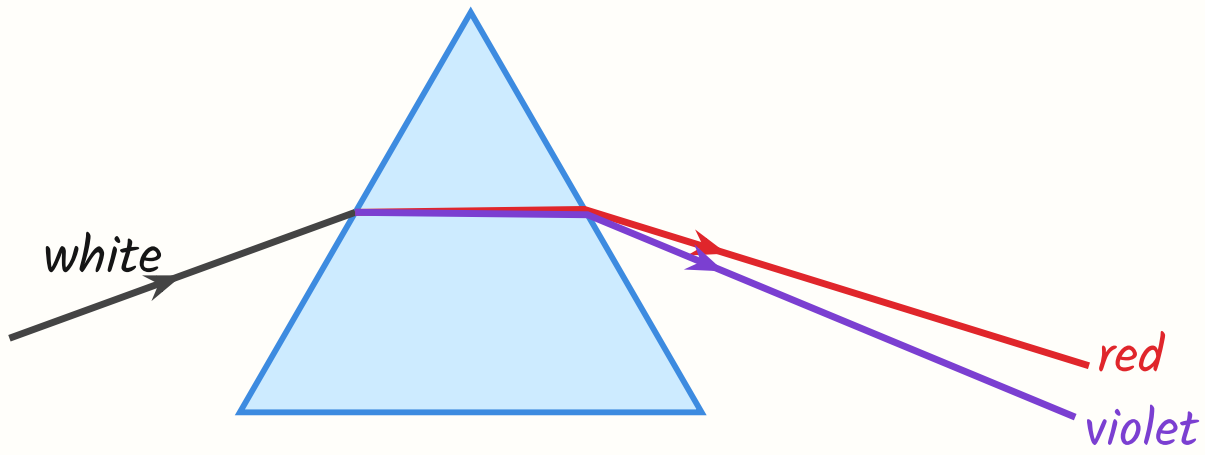
- *Grazing incidence ($i = 90°$): $r_1 = C$, deviation MAXIMUM (same at grazing emergence).*
- *Emerges only if $r_2 \le C$: $i \ge i_{min}$, $\sin i_{min} = n \sin(A - C)$ ($A > C$).*
- *NO ray emerges for any $i$ if $A > 2C$.*

**Example**

$A = 60°$, $n = 1.5$ ($C = 41.8°$): $i_{min} = 27.9°$; $\delta_{max} = 90° + 27.9° - 60° = 57.9°$.

# Dispersion and Scattering

## 1 Dispersion by a prism




*traced exactly,* $n_r = 1.50$, $n_v = 1.56$ *(spread exaggerated)*

$$\theta = \delta_v - \delta_r = (n_v - n_r)A, \qquad \omega = \frac{n_v - n_r}{n_y - 1} = \frac{\theta}{\delta_y}$$

*THIN prism, small* $i$ · $\theta$*: angular dispersion* · $\omega$*: dispersive power (material only, not* $A$*)* · $\delta_y = (n_y - 1)A$*: mean deviation*

- $n$ *depends on colour (Cauchy:* $n = a + b/\lambda^2$*), larger for violet: violet deviates MOST, red LEAST.*

**Example**

*Crown:* $n_r = 1.513$, $n_y = 1.517$, $n_v = 1.523$, $A = 10°$: $\theta = 0.1°$, $\delta_y = 5.17°$, $\omega = \frac{0.010}{0.517} = 0.0193$.

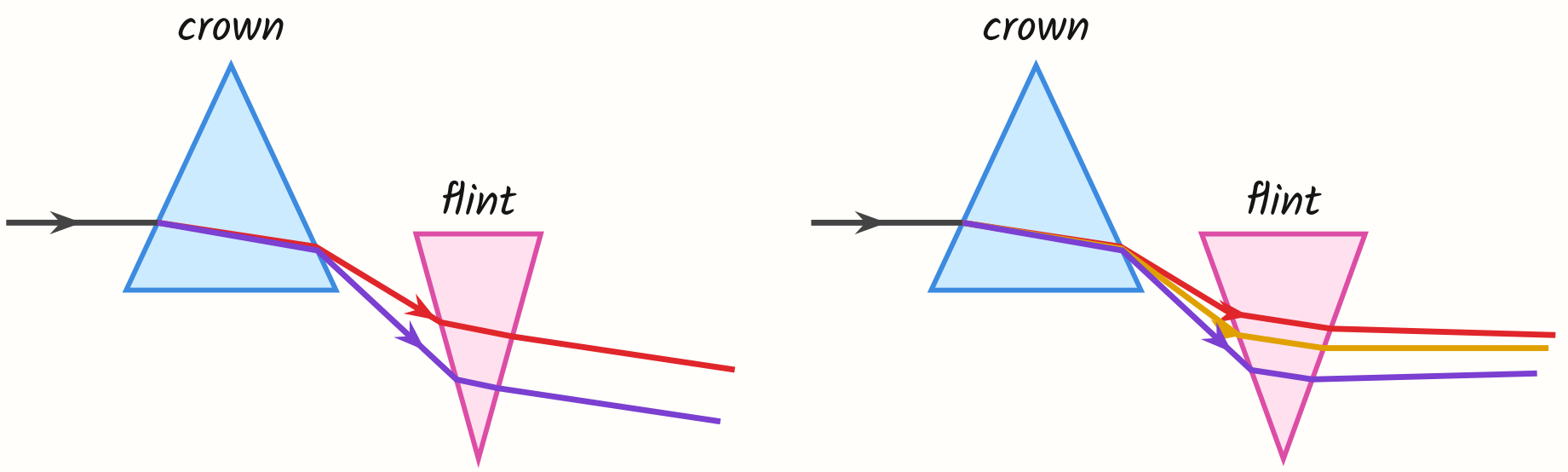
## 2 Scattering (basics)

$$I_{scattered} \propto \frac{1}{\lambda^4}$$

*Rayleigh: particles much SMALLER than* $\lambda$ *(air molecules)* · *blue sky; red Sun at sunrise and sunset (long path scatters the blue away)* · *white clouds: droplets* $\gg \lambda$ *scatter all colours almost equally*

# Achromatic and Direct Vision

## 1 Two prisms, one inverted




ACHROMATIC: red ‖ violet — DIRECT VISION: yellow undeviated

exact trace, $n$ exaggerated (r/y/v): crown $1.50/1.56/1.62$, flint $1.62/1.74/1.86$

## 2 Achromatic: deviation without dispersion

$$(n_v - n_r)A + (n_v' - n_r')A' = 0 \;\Leftrightarrow\; \omega\delta + \omega'\delta' = 0$$

THIN prisms, the second INVERTED ($A'$, $\delta'$ negative) · net $\delta + \delta' = \delta\left(1 - \frac{\omega}{\omega'}\right)$ · lens doublet in contact: $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$

## 3 Direct vision: dispersion without deviation

$$(n_y - 1)A + (n_y' - 1)A' = 0$$

thin prisms, second inverted · mean (yellow) ray undeviated, net dispersion $\delta(\omega - \omega') \neq 0$

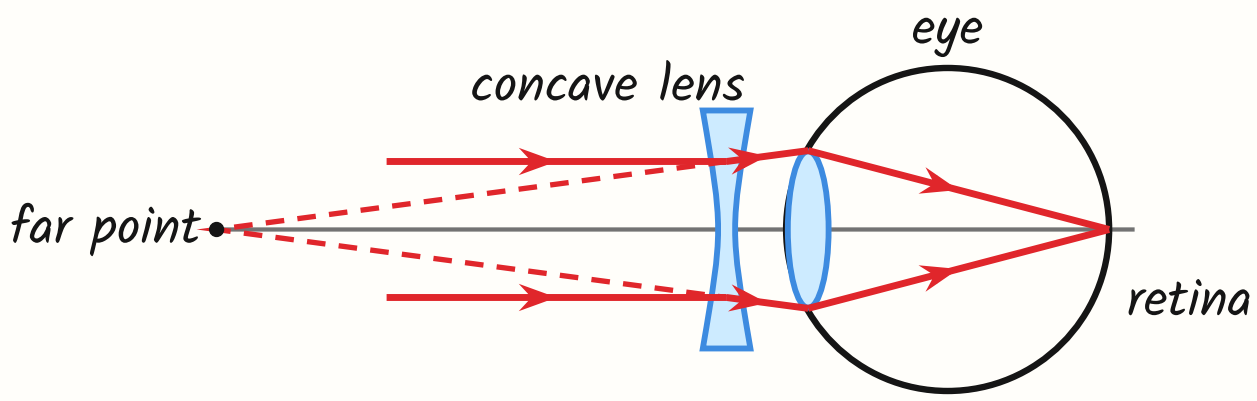
**Example**

Crown $A = 6°$, $n_y = 1.5$, $\omega = 0.02$ ($\delta = 3°$); flint $n_y' = 1.6$, $\omega' = 0.03$. Achromatic: $\delta' = -2°$, $A' = 3.33°$, net $1°$. Direct vision: $A' = 5°$, net dispersion $-0.03°$.

# Human Eye and its Defects

*normal eye: near point $25\,\mathrm{cm}$, far point $\infty$; lenses close to the eye*

## 1 Myopia (short sight)




- *Far point at FINITE $x$: a distant object focuses IN FRONT of the retina.*
- *CONCAVE lens $f = -x$ $(P = -1/x)$: parallel rays seem to come from the far point. Far point $2\,\mathrm{m}$: $P = -0.5\,\mathrm{D}$.*

## 2 Hypermetropia (long sight)

$$\frac{1}{f} = \frac{1}{25} - \frac{1}{y} \quad (cm)$$

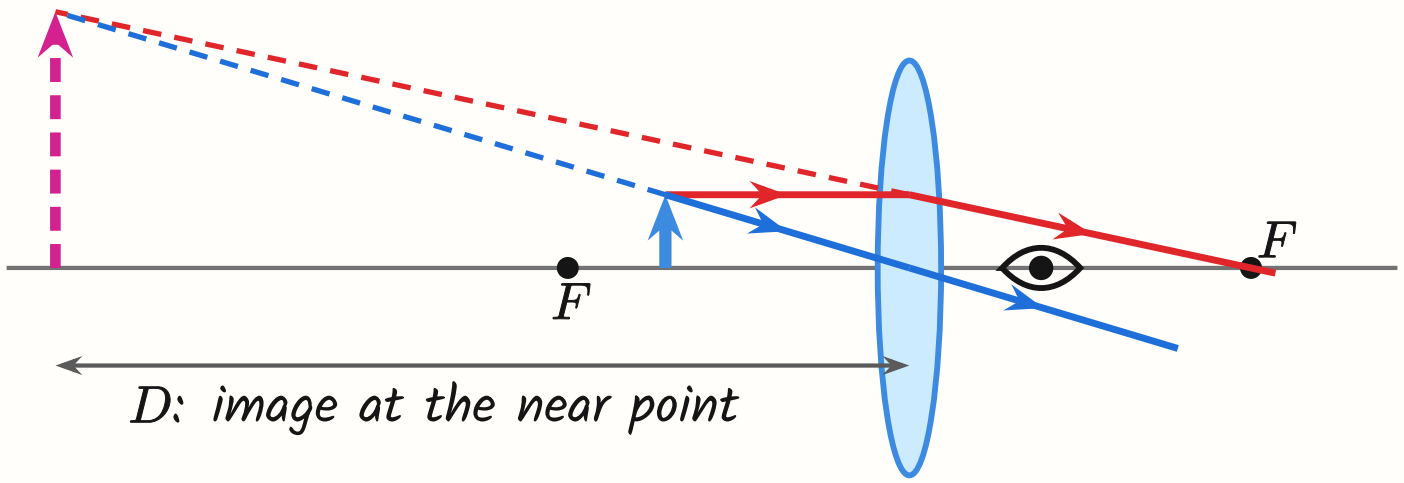
*near point at $y > 25\,\mathrm{cm}$ · CONVEX lens: object at $25\,\mathrm{cm}$ appears at $y$ · $y = 50\,\mathrm{cm}$: $f = +50\,\mathrm{cm}$, $P = +2\,\mathrm{D}$*

## 3 Accommodation, presbyopia, astigmatism

- *Ciliary muscles change the eye-lens $f$; power of accommodation $\frac{1}{d_{near}} - \frac{1}{d_{far}}$ $(m) \approx 4\,\mathrm{D}$ (normal young eye).*
- *Presbyopia (age): bifocals. Astigmatism (cornea curved unequally): CYLINDRICAL lens.*

# Simple Microscope

## 1 Magnifying glass




$$M = \frac{\beta}{\alpha} = 1 + \frac{D}{f} \quad \text{(image at } D\text{)}$$

$$M = \frac{D}{f} \quad \text{(image at } \infty\text{, relaxed eye)}$$

thin convex lens held CLOSE to the eye, small angles · $\alpha = h/D$: angle of the object at the near point, naked eye · $\beta$: angle of the image · $D = 25\,\text{cm}$

## 2 Remember

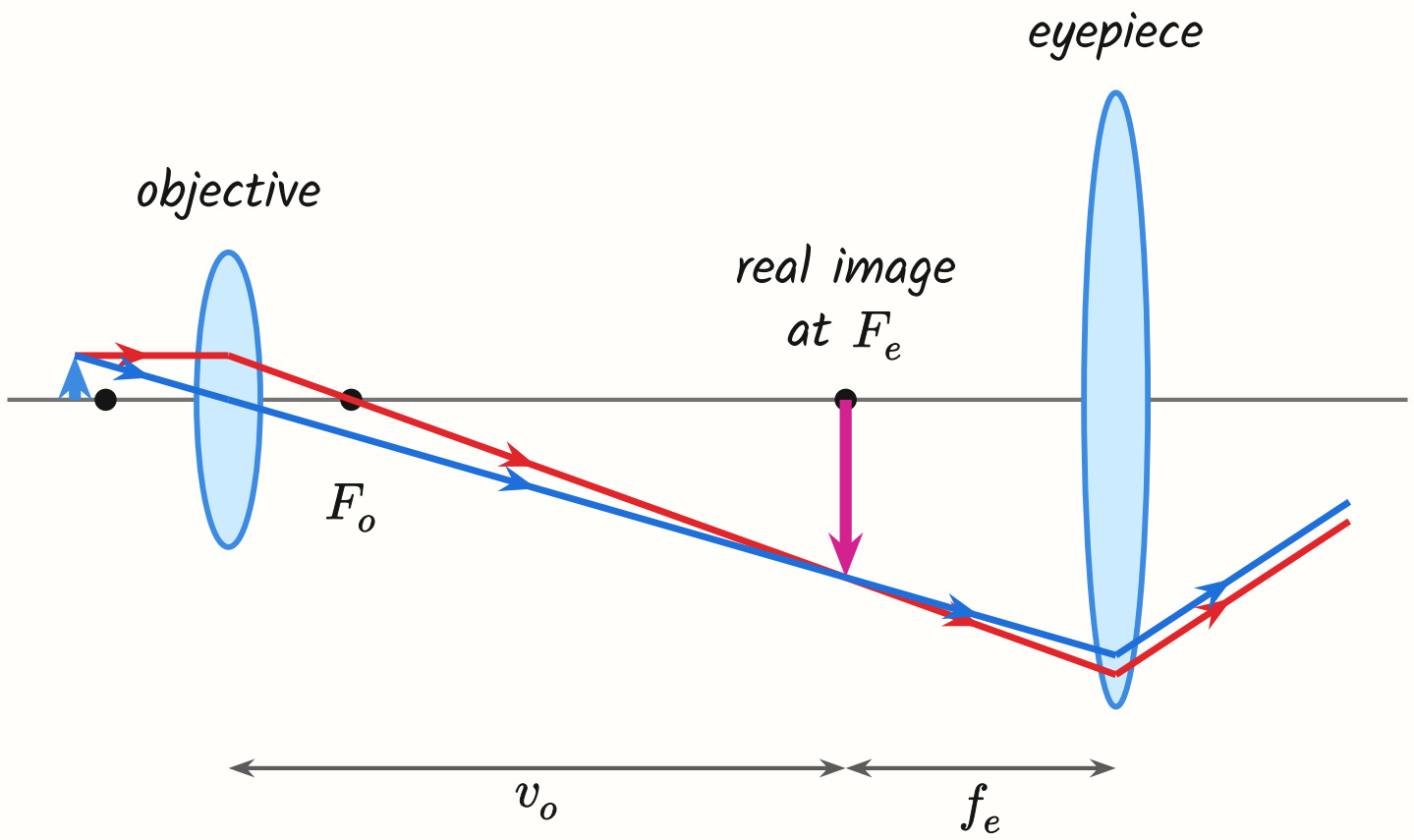
- Image at $D$: object just inside $F$, $M$ is largest; image at $\infty$: object AT $F$, eye relaxed (normal use).
- $M$ is ANGULAR magnification; only for the image at $D$ does it equal the linear $m = v/u = 1 + D/f$.
- Useful only if $f < D$.

**Example**

$f = 5\,\text{cm}$: $M = 1 + \frac{25}{5} = 6$ (image at $D$); $M = \frac{25}{5} = 5$ (relaxed eye).

# Compound Microscope

## 1 Ray diagram (normal adjustment)




$$|M| = \frac{v_o}{|u_o|} \cdot \frac{D}{f_e} \quad \text{(final image at } \infty\text{)}$$

$$|M| = \frac{v_o}{|u_o|}\left(1 + \frac{D}{f_e}\right) \quad \text{(final image at } D\text{)}$$

*thin lenses, paraxial, eye close to the eyepiece · final image inverted and virtual · length $v_o + f_e$ (at $\infty$) or $v_o + |u_e|$ (at $D$)*

- *Object just outside $F_o$, intermediate image near $F_e$:* $|M| \approx \dfrac{L}{f_o} \cdot \dfrac{D}{f_e}$

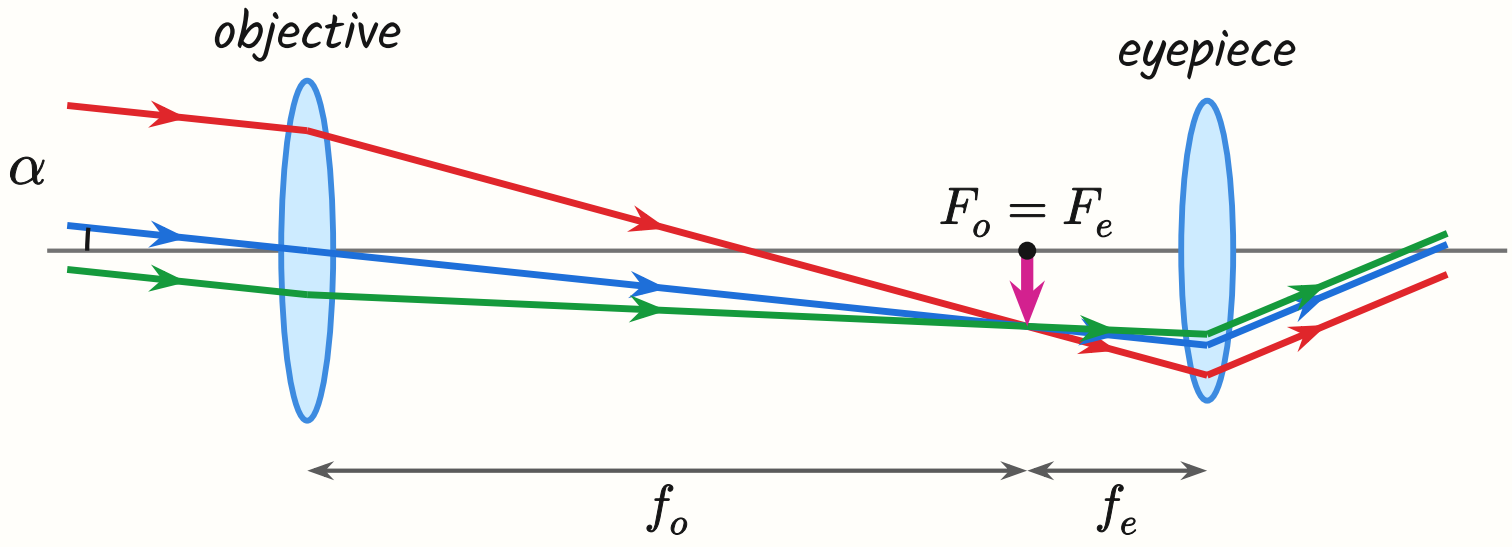
  *($L$: tube length, from $F_o'$ to $F_e$). Small $f_o$ and $f_e$ give large $M$.*

**Example**

$f_o = 2\,\text{cm}$, $u_o = -2.5\,\text{cm}$: $v_o = 10\,\text{cm}$, $m_o = -4$; $f_e = 5\,\text{cm}$,
*relaxed eye:* $|M| = 4 \times \frac{25}{5} = 20$, *length* $15\,\text{cm}$.

# Astronomical Telescope

## 1 Ray diagram (normal adjustment)




$$M = \frac{\beta}{\alpha} = \frac{f_o}{f_e}, \qquad L = f_o + f_e$$

NORMAL adjustment: object and final image at infinity · thin lenses, small angles · image inverted

$$M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right), \qquad L = f_o + \frac{f_e D}{f_e + D}$$

final image at the near point $D = 25\,\text{cm}$

**Example**

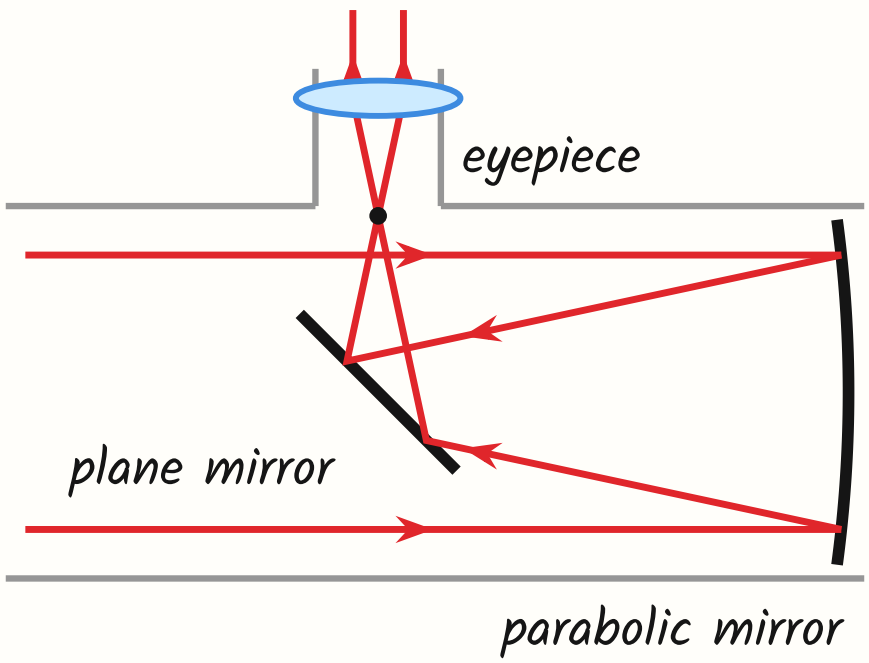
$f_o = 150\,\text{cm}$, $f_e = 5\,\text{cm}$: normal $M = 30$, $L = 155\,\text{cm}$; image at $D$: $M = 30\left(1 + \frac{5}{25}\right) = 36$.

## 2 Upright images (normal adjustment)

- Terrestrial: erecting lens ($f$) in between: $L = f_o + f_e + 4f$, $M = f_o/f_e$, erect.
- Galilean: CONCAVE eyepiece: $L = f_o - |f_e|$, $M = f_o/|f_e|$, erect.

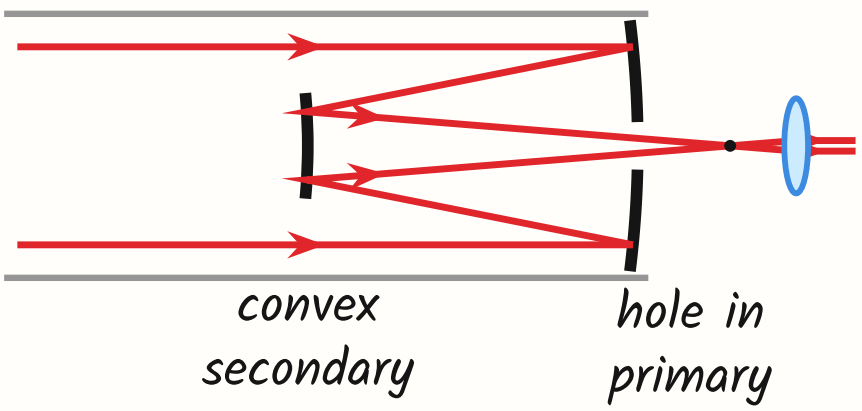
# Reflecting Telescopes

## 1 Newtonian telescope (rays traced exactly)




- Parabolic primary sends ALL rays parallel to its axis to its focus (no spherical aberration).
- A plane mirror at $45°$ turns the beam sideways to the eyepiece.

## 2 Cassegrain telescope




- Convex (hyperbolic) secondary sends light back through a HOLE in the primary: short tube, long focal length.

## 3 Why mirrors?

- No chromatic aberration; parabolic: no spherical aberration.
- Supported from the back, cheaper: large aperture, more light, better resolution.
- $M = f_o/f_e$ (normal adjustment, paraxial); Newtonian: $f_o = R/2$ of the primary; Cassegrain: $f_o =$ effective focal length of primary + secondary (longer than $R/2$).

# Common Traps

## 1 Exam traps

**Avoid these**

- ✗ $f = R/2$ for every ray (only PARAXIAL rays; wide rays focus closer).
- ✗ A mirror's $f$ changes in water (no: $f = R/2$, independent of the medium; a LENS's $f$ does change).
- ✗ Apparent depth $h/n$ at any viewing angle (only NEAR-NORMAL viewing).
- ✗ TIR from air into glass (only DENSER to RARER, and only for $i > C$).
- ✗ Lateral shift $= t(1 - 1/n)$ (that is the NORMAL shift; lateral shift is $t \sin(i - r)/\cos r$ for a parallel slab).
- ✗ $\delta = (n - 1)A$ for any prism (only THIN prism, small $i$).
- ✗ Telescope $M = f_o/f_e$ always (only NORMAL adjustment; image at $D$: $\frac{f_o}{f_e}(1 + \frac{f_e}{D})$).
- ✗ Convex lens always converges (only if $n_{lens} > n_{medium}$).
- ✗ $v_I = -m^2 v_O$ for a lens too (lens: $v_I = +m^2 v_O$; both along the axis).
- ✗ Taking $u > 0$ for a real object (Cartesian, light from the left: real object $u < 0$).
- ✗ Dispersive power depends on $A$ (no: material only, thin prism).

**Golden points**

- ★ First ask: paraxial? thin lens? which medium? final image at $\infty$ or at $D$?

# Standard Results

## 1 One-line results

| Quantity | Result | Condition |
|---|---|---|
| Mirror | $\frac{1}{v}+\frac{1}{u}=\frac{1}{f},\ m=-\frac{v}{u}$ | paraxial, Cartesian |
| Lens | $\frac{1}{v}-\frac{1}{u}=\frac{1}{f},\ m=\frac{v}{u}$ | thin, paraxial, Cartesian, same medium both sides |
| Lensmaker | $\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$ | thin lens in air |
| Apparent depth | $h/n$ | near-normal viewing |
| Lateral shift | $t\sin(i-r)/\cos r$ | parallel slab |
| Critical angle | $\sin C=1/n$ | denser into air |
| Prism | $n=\frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$ | prism in air, minimum deviation |
| Thin prism | $\delta=(n-1)A$ | small $A$, small $i$ |
| Microscope | $\frac{L}{f_o}\cdot\frac{D}{f_e}$ | approx., image at $\infty$ |
| Telescope | $f_o/f_e,\ L=f_o+f_e$ | normal adjustment |

## 2 Numbers to remember

- Critical angle: glass $(1.5)$ $41.8°$, water $(4/3)$ $48.6°$. Thin glass $(1.5)$ lens in water: $f_w=4f_a$.
- Silvered plano-convex: $R/[2(n-1)]$ (plane face) or $R/(2n)$ (curved face). Equilateral glass prism $(1.5)$: $\delta_m=37.2°$.

**Summary**

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ • $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ • $n_1\sin i=n_2\sin r$ • $P=P_1+P_2$

mirror, thin lens: paraxial, Cartesian · $P_1+P_2$: thin lenses in contact

# Questions: JEE Main Level (I)

*paraxial rays, thin lenses in air, Cartesian sign convention*

**Q1 · JEE Main level · MCQ**

Object $15\,\text{cm}$ in front of a concave mirror, $|f| = 10\,\text{cm}$. Image: (A) real, inverted, $30\,\text{cm}$ in front, $2\times$ (B) virtual, erect, $30\,\text{cm}$ behind (C) real, inverted, $6\,\text{cm}$ in front (D) virtual, erect, $6\,\text{cm}$ behind

Solution: $u = -15$, $f = -10$: $\frac{1}{v} = -\frac{1}{10} + \frac{1}{15} = -\frac{1}{30}$, $v = -30\,\text{cm}$ (real, in front); $m = -\frac{v}{u} = -2$.

**Answer: (A)**

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**Q2 · JEE Main level · Numerical**

A coin lies under $20\,\text{cm}$ of water ($n = 4/3$). Seen from directly above (near-normal), by how many cm does it appear raised?

Solution: $h' = \frac{20}{4/3} = 15\,\text{cm}$; rise $= 20 - 15 = 5\,\text{cm}$.

**Answer: 5**

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**Q3 · JEE Main level · MCQ**

Light travels at $2 \times 10^8\,\text{m s}^{-1}$ in a medium. Critical angle, medium to air: (A) $\sin^{-1}(2/3)$ (B) $\sin^{-1}(1/3)$ (C) $\sin^{-1}(3/4)$ (D) $\sin^{-1}(1/2)$

Solution: $n = \frac{3\times10^8}{2\times10^8} = 1.5$; $\sin C = \frac{1}{n} = \frac{2}{3}$, $C \approx 41.8°$.

**Answer: (A)**

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**Q4 · JEE Main level · Numerical**

A thin equi-biconvex glass lens ($n = 1.5$, $|R| = 20\,\text{cm}$) has $f = 20\,\text{cm}$ in air. Its focal length (in cm) in water ($n = 4/3$)?

Solution: $\frac{1}{f_w} = \left(\frac{1.5}{4/3} - 1\right)\left(\frac{1}{20} + \frac{1}{20}\right) = \frac{1}{8} \cdot \frac{1}{10} = \frac{1}{80}$.

**Answer: 80**

# Questions: JEE Main Level (2)

*paraxial rays, thin lenses in air, $D = 25\,\text{cm}$*

**Q5 · JEE Main level · MCQ**

*Thin lenses of $+20\,\text{cm}$ and $-30\,\text{cm}$ in contact. The pair is: (A) converging, $60\,\text{cm}$ (B) diverging, $60\,\text{cm}$ (C) converging, $12\,\text{cm}$ (D) diverging, $12\,\text{cm}$*

*Solution:* $P = \frac{1}{0.2} - \frac{1}{0.3} = \frac{5}{3}\,\text{D} > 0$; $F = 0.6\,\text{m} = 60\,\text{cm}$, *converging.*

**Answer: (A)**

**Q6 · JEE Main level · Numerical**

*A prism of angle $60^\circ$ (in air) gives a minimum deviation of $30^\circ$. Its refractive index?*

*Solution:* $n = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2} \approx 1.41$.

**Answer:** $\approx 1.41$

**Q7 · JEE Main level · Numerical**

*Compound microscope: $f_o = 1\,\text{cm}$, $f_e = 5\,\text{cm}$, object $1.1\,\text{cm}$ from the objective, final image at the near point. $|M|$?*

*Solution:* $\frac{1}{v_o} = \frac{1}{1} - \frac{1}{1.1} = \frac{1}{11}$, $v_o = 11\,\text{cm}$, $|m_o| = 10$; $m_e = 1 + \frac{25}{5} = 6$; $|M| = 10 \times 6 = 60$.

**Answer:** 60

**Q8 · JEE Main level · MCQ**

*Astronomical telescope, $f_o = 100\,\text{cm}$, $f_e = 5\,\text{cm}$, normal adjustment. $M$ and length: (A) 20, $105\,\text{cm}$ (B) 24, $105\,\text{cm}$ (C) 20, $95\,\text{cm}$ (D) 24, $104.2\,\text{cm}$*

*Solution:* $M = \frac{100}{5} = 20$, $L = 100 + 5 = 105\,\text{cm}$ *(image at $D$ would give 24: trap).*

**Answer: (A)**

# Questions: JEE Advanced Level (I)

*paraxial rays, thin lenses in air, Cartesian sign convention*

**A1 · JEE Advanced level · Numerical**

A thin plano-convex lens ($n = 1.5$, $R = 20\,\text{cm}$) has its PLANE face silvered. An object is placed on its axis $30\,\text{cm}$ in front of it. How far (in cm) from the lens is the image?

Solution: $P_{eq} = 2P_L + P_M = 2 \cdot \frac{0.5}{0.2} + 0 = 5\,\text{D}$, $f_{eq} = -20\,\text{cm}$ (concave mirror). $\frac{1}{v} = -\frac{1}{20} + \frac{1}{30} = -\frac{1}{60}$; $v = -60\,\text{cm}$, real, $m = -2$.

**Answer:** 60

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**A2 · JEE Advanced level · Multi-correct**

A glass sphere ($n = 1.5$) of radius $10\,\text{cm}$ is in air. A point object is on a diameter line, $30\,\text{cm}$ from the near surface. Then: (A) the near surface alone images it $90\,\text{cm}$ beyond that surface (B) the final image is real, $14\,\text{cm}$ beyond the far surface (C) the final image is virtual (D) the first image lies inside the sphere

Solution: Surface 1: $\frac{1.5}{v} - \frac{1}{-30} = \frac{0.5}{10}$, $v = 90\,\text{cm}$ (outside the $20\,\text{cm}$ sphere). Surface 2: $u = 90 - 20 = +70\,\text{cm}$, $\frac{1}{v} - \frac{1.5}{70} = \frac{-0.5}{-10}$, $v = 14\,\text{cm}$ (real).

**Answer:** (A), (B)

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**Golden points**

- ★ Silvered lens: light crosses the lens twice, $P_{eq} = 2P_L + P_M$; then use the MIRROR formula with $f_{eq} = -1/P_{eq}$.
- ★ Two surfaces: the first image is the object for the second; re-measure from the new vertex ($90 - 20 = 70\,\text{cm}$, a virtual object).

# Questions: JEE Advanced Level (2)

## paraxial rays, thin lenses in air

### A3 · JEE Advanced level · Multi-correct

A convex lens ($f = 20\,\text{cm}$) has a plane mirror $20\,\text{cm}$ behind it, perpendicular to the axis. An object is $60\,\text{cm}$ in front of the lens. Then: (A) the lens first images it $30\,\text{cm}$ behind the lens (B) the final image is $20\,\text{cm}$ from the lens, at the mirror (C) the final image is real (D) the final image is inverted and of the same size

Solution: Lens: $\frac{1}{v} = \frac{1}{20} - \frac{1}{60}$, $v = 30$, $m_1 = -0.5$. That is $10\,\text{cm}$ behind the mirror; the mirror images it $10\,\text{cm}$ in front (at $10\,\text{cm}$ from the lens), $m_2 = 1$. Light now goes right to left: $u = -10$, $\frac{1}{v} = \frac{1}{20} - \frac{1}{10}$, $v = -20$: virtual, $20\,\text{cm}$ from the lens on the mirror side; $m_3 = 2$. Total $m = -1$.

**Answer: (A), (B), (D)**

### A4 · JEE Advanced level · Paragraph

An optical fibre has core index $1.5$ and cladding index $1.2$. (i) Largest angle of incidence at the end face (from air) for which light is guided? (ii) The same if the end face is in water ($4/3$)?

Solution: (i) $\sin\theta_a = \sqrt{1.5^2 - 1.2^2} = 0.9$, $\theta_a = 64.2^\circ$. (ii) $\frac{4}{3}\sin\theta_a = 0.9$, $\sin\theta_a = 0.675$, $\theta_a = 42.5^\circ$.

**Answer: (i) $64.2^\circ$ (ii) $42.5^\circ$**

### Golden points

- ★ After a mirror the light travels right to left: take $+$ along the NEW direction of the light before using the lens formula again.
- ★ Fibre: $n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ is fixed, so a denser outside medium gives a smaller acceptance angle.

# Questions: JEE Advanced Level (3)

## real object, paraxial rays, lens or mirror in air

**AS · JEE Advanced level · Matching**

List I: (P) convex mirror, object anywhere (Q) concave mirror, object between $F$ and $P$ (R) convex lens, object at $2F_1$ (S) convex lens, object at $F_1$. List II: (1) virtual, erect, diminished (2) virtual, erect, magnified (3) real, inverted, same size (4) image at infinity

Solution: (P) always virtual, erect, diminished; (Q) $|u| < |f|$ gives $v > 0$, $m > 1$; (R) $v = 2f$, $m = -1$; (S) $\frac{1}{v} = \frac{1}{f} - \frac{1}{f} = 0$.

**Answer: P-1, Q-2, R-3, S-4**

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**A6 · JEE Advanced level · Numerical**

An object on the axis of a concave mirror ($|f| = 20\,\mathrm{cm}$), $30\,\mathrm{cm}$ away, moves towards the mirror at $2\,\mathrm{cm\,s^{-1}}$ (mirror at rest). Speed of the image in $\mathrm{cm\,s^{-1}}$?

Solution: $\frac{1}{v} = -\frac{1}{20} + \frac{1}{30}$, $v = -60\,\mathrm{cm}$, $m = -2$.

$v_I = -m^2 v_O = -4 \times (+2) = -8\,\mathrm{cm\,s^{-1}}$: the image moves AWAY from the mirror at $8\,\mathrm{cm\,s^{-1}}$.

**Answer:** 8

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**Golden points**

- ★ Image velocity: mirror $v_I = -m^2 v_O$, lens $v_I = +m^2 v_O$ (along the axis).
- ★ Nature from the sign of $m$ (single mirror or lens, real object): $m < 0$ real and inverted, $m > 0$ virtual and erect.